

Table 4. AvgCurrDet Threshold when using 10mΩ and Default nProtMiscTh.CurrDet = 7.5mA

	AVGCURRENT FILTER CONFIGURATION (nFilterCfg.nCurr)							
	1 (0.7s)	2 (1.4s)	3 (2.8s)	4 (5.6s)	5 (11.25s)	6 (22.5s)	7 (45s)	8 (90s)
Active (0.351s)	4.22mA	2.34mA	2.34mA	1.41mA (default)	1.41mA	0.94mA	0.94mA	0.7mA
Hibernate (1.4s)	7.5mA	4.2mA	4.2mA	2.3mA	2.3mA	1.4mA	1.4mA	0.94mA
Hibernate (2.8s)	7.5mA	7.5mA	7.5mA	4.2mA	4.2mA	2.3mA	2.3mA	1.4mA

The fast responses in [Table 4](#) correspond with the 0.351s ADC update rate. The more accurate slow responses correspond with the AvgCurrent filter delay configuration.

Protector

Simple protection schemes are available to protect a battery from exceeding the safe levels. These schemes include protection for overdischarge current, short-circuit current, overcharge current, undervoltage and overvoltage. The next level of protection offers smart protection schemes which include protection for under OCV (SmartEmpty), long overdischarge current, overtemperature limits for charge and discharge, undertemperature charge limits, and charge-done protection. The MAX17335 provides all of these simple and smart protection schemes with programmable thresholds and programmable timer delays for each fault.

The MAX17335 provides additional protection functionality beyond these schemes as follows:

Discharging Protection Functionality

- **Overcurrent:** (see [nODSCCfg](#) and nODSCTh)
 - **Fast Short-Circuit (70μs to 985μs):** The short-circuit comparator is programmable from 5.12mV to 158.72mV with delay programmable from 70μs to 985μs.
 - **Medium (1ms to 15ms):** The overdischarge current comparator is programmable from 2.55mV to 79.36mV with delay programmable from 1ms to 15ms.
 - **Slow (351ms to 35s):** Slow overdischarge protection is programmable from 0mV to 51.2mV in 0.2mV steps with delay programmable from 351ms to 35s (see nDelayCfg and [nIPrtTh1](#)).
- **Overtemperature:**
 - **Hot (OTPD—Overtemperature Discharge):** Discharge overtemperature (OTPD, see [nProtMiscTh](#)) is separately programmable from charge overtemperature (OTPC). OTPD is typically a higher temperature than OTPC, since charging while hot is more hazardous than discharging. OTPD is programmable in 1°C steps, with a programmable timer (see nDelayCfg).
 - **Die-Hot:** The MAX17335 measures die temperature as well as a thermistor's temperature. Since the IC is generally located close to the external FETs, the die temperature can indicate when the FETs are overheating. This separately programmable threshold (see [nProtMiscTh](#)) blocks both charging and discharging.
 - **Permanent-Fail-Hot:** When a severe overtemperature is detected, the fault is recorded into NVM and permanently disables the charge and discharge FETs and blows the three terminal fuse if enabled.
- **Too Cold Discharge:** If enabled, the IC blocks discharging if the cell temperature is too low. It prevents the discharge of a cell when cell impedance due to temperature is too large to support the application load.
- **Undervoltage (UVP):** Undervoltage is protected by three thresholds: UVP (undervoltage protect), UVShdn (undervoltage shutdown), and UOCVP (under OCV protect—[SmartEmpty](#)). UOCVP provides deep-discharge-state protection that is immune from load and cell impedance/resistance variations.

Charging Protection Functionality:

- **Overvoltage Protection (OVP):** Overvoltage protection is programmable with 10mV resolution (see nOVPrTh). Temperature-region-dependent OVP protection is also provided for cold/room/warm and hot temperature regions (see [nVChgCfg1/2](#)). OVP detection is debounced with a programmable timer (see nDelayCfg). An additional, higher OVP permanent failure threshold is programmable, which records any excessive OVP into NVM and permanently blocks charging.
- **Charge Temperature Protection:** Temperature protection thresholds are debounced with a programmable timer (see

nDelayCfg).

- **Hot (OTPC):** Charging temperature protection is programmable with 2.5°C resolution (see [nTPrtTh1](#)) and 2.5°C hysteresis.
- **Cold (UTP):** Charging is blocked at cold, programmable with 2.5°C resolution (see [nTPrtTh1](#)), and no hysteresis.
- **Overcharge-Current Protection:**
 - **Fast:** Overcharge current is detected by a programmable hardware comparator and debounce timer between 0 to 39.375mV and 1ms to 15ms thresholds.
 - **Slow:** A lower and slower overcharge current protection ensures that more moderate high currents do not persist for a long time. With a 10mΩ sense resistor, this is programmable up to 5.12A in 40mA steps, with an additional delay programmable between 0.35s and 22.5s. Additionally, with [nNVCfg1.enJP](#) = 1, this overcurrent protection threshold is modulated according to temperature region (see [nIChgCfg1/2](#)).
- **Charge-Done:** If enabled, the IC blocks charge whenever charge termination is detected, until discharging or charger removal is eventually detected.
- **Charger-Communication Timeout:** If enabled, during charging the IC turns off the charge FET if the host has stopped communicating beyond a timeout configurable from 11s to 3 minutes. In systems that consult the battery for prescribing the charge current or charge voltage, especially to apply JEITA thresholds or step-charging, this feature is useful to protect against operating system crash or shutdown.
- **Overcharge-Capacity Alert:** If any charge session delivers more charge (coulombs) to the battery than the expected full design capacity, charging is briefly interrupted and a protection alert is indicated. This threshold is programmable as a percentage (see [nProtMiscTh.QOvflwTh](#)) beyond the design capacity.

Other Faults:

- **Nonvolatile CheckSum Failure:** If enabled ([nNVCfg1.enProtChkSm](#)), the MAX17335 blocks charge and discharge when startup checksum of protector NVM does not match the value stored in [nProtCfg2.CheckSum](#).

Other Protection Functionality:

- **Zero-Volt Charging:** The IC can begin charging when the cell has depleted to 2.16V (ZVC disabled) or from 0.0V (ZVC enabled). See the [Zero-Volt Charging](#) section for more details.
- **Overdischarge-Removal Detection:** Following any overdischarge current fault, after the IC turns off the discharge FET, it tests for load removal by sourcing 30μA into PCKP. Load removal is detected when PCKP exceeds 1V. This low threshold is intentionally below the startup voltage of most ICs in order to allow active loads by external ICs while rejecting passive loads by resistors (short-circuit, failed components, etc.).
- **Charger Removal Detection:** Following any charge fault, after the IC turns off the charge FET, it measures PCKP to detect the removal of the offending charger. Charger removal is detected when PCKP falls below [BATT - nIPrtTh2.ChgDetTh](#) or whenever discharge current is detected.
- **Battery Internal Self-Discharge Detection:** The IC measures the internal self-discharge of the battery that might indicate health or safety problems. The IC alerts the system or turns off the charge and discharge FETs when a leakage is detected above the configurable threshold. See the [Battery Internal Self-Discharge](#) section for more details.
- **Ideal-Diode Control:** During any charge fault, the charge FET turns on when a discharge current is detected. See the [Ideal Diode](#) section for more details. The discharge FET behaves the same way during discharge faults to block discharging, yet turns on during charging. This ideal diode behavior reduces the heat and voltage drop associated with the body diode during protection faults.

Protection Fault Reporting:

- **Protection Fault Status:** Each charge and discharge fault state is latched in the ProtStatus register. When the fault is cleared, the corresponding bit is cleared.
- **Protection Fault Alerts:** The ProtAlrt register latches the status of any previous faults detected by the device. Once a fault is detected, the corresponding bit remains set until it is cleared by the host. Additionally, the Status.ProtAlrt bit is set when any ProtAlrt bit is set.
- **Protection Fault Logging:** The nFaultLog register also indicates which protection events happened during each history log period.

Charging Regulation Registers: The [ChargingVoltage](#) and [ChargingCurrent](#) registers control and display the calculated target charging voltage and current. This includes the following knowledge which generally is associated with a particular battery and can be stored in the battery with the MAX17335.

- **Factory Recommended Charging Current and Voltage:** This is useful when a system involves multiple battery vendors, swappable batteries, aftermarket batteries, or legacy system support.
- **Manual Charge Control:** If enabled, the host can manually override the autonomous control and write to the ChargingVoltage and ChargingCurrent registers directly.
- **Charging Modifications According to Battery Temperature and Age:** Significantly above and below room temperature, most cell manufacturers recommend charging at reduced current and lower termination voltage to assure safety and improve lifespan. The MAX17335 modulates its settings according to TooCold/Cold2/Cold1/Cool/Room/Warm/Hot1/Hot2/TooHot programmable temperature regions (see [nTPrtTh1/2](#)). Both charging current and voltage are modulated at Cold2/Cold1/Cold/Warm/Hot1/Hot2, targeting lower than Room (see [nVChgCfg1/2](#) and [nIChgCfg1/2](#)). If enabled, the MAX17335 can derate the ChargingVoltage and ChargingCurrent as the battery ages. See [nAgeChgCfg](#).
- **Step-Charging:** A common practice to balance lifespan and charge speed is to apply step-charging profiles (see the [Step-Charging](#) section). The MAX17335 supports five programmable steps with programmable charge currents and voltages.

At a high level, the MAX17335 protector has a state machine as shown in [Figure 8](#). Each charge and discharge fault state is latched in the [ProtStatus](#) register, where each fault obeys a separate instance of the state machine shown in [Figure 8](#). Any single charge fault opens the charge FET to block charge current (charge faults are OR'd together). All charge faults must be released to allow the charge to resume (charge fault release conditions are AND'd together). The behavior is similar for blocking discharge.

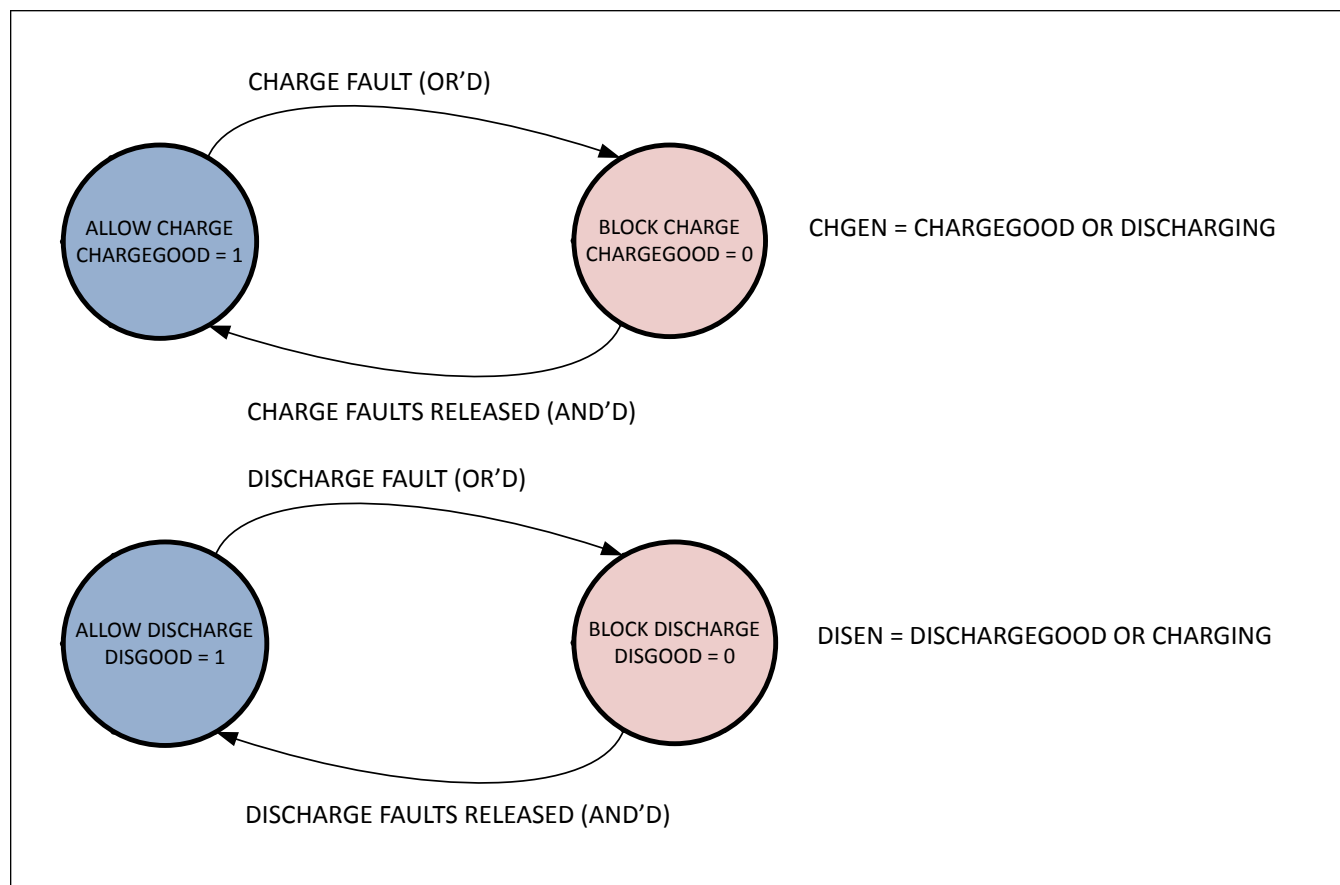


Figure 8. Simplified Protector State Machine

The IC includes a write protection and a permanent locking function. The write protection prevents accidental overwrites of protection parameters. This protection must be cleared before updating any registers and should be set after

configuration changes are made. The permanent locks prevent intentional or malicious tampering and should be enabled after development is completed and the battery pack is ready to ship in production. See the [Memory Locks and Write Protection](#) section for more details.

The protector registers are summarized by their protection function in [Table 5](#) and are graphically shown across the various temperature ranges in [Figure 9](#) and [Figure 10](#).

Table 5. Summary of Protector Registers by Function

FUNCTION	REGISTER
VOLTAGE THRESHOLDS	
Permanent Fail Overvoltage Protection	nOVPrTh
Overvoltage Protection	nVChgCfg1/2, nOVPrTh
Overvoltage Protection Release	nOVPrTh
UnderOCV Protection	nUVPrtTh
Undervoltage Protection	nUVPrtTh
Undervoltage Shutdown	nUVPrtTh
Prequalification Voltage	nChgCfg0
CURRENT THRESHOLDS	
Fast Overcharge Protection	nODSCTh, nODSCCfg
Slow Overcharge Protection	nIPrtTh1
Slow Overdischarge Protection	nIPrtTh2
Fast Overdischarge Protection	nODSCTh, nODSCCfg
Short-Circuit Protection	nODSCTh, nODSCCfg
Charging Detected	nProtMiscTh
Discharging Detected	nProtMiscTh
Temperature Thresholds	nTPrtTh1, nTPrtTh2, nProtMiscTh
Fault Timers	nDelayCfg
CHARGING REGULATION	
Charging Voltage	nVChgCfg1, nVChgCfg2
Charging Current	nIChgCfg1, nIChgCfg2
Precharge Current	nChgCfg0
Step Charging	nStepCurr, nStepVolt
Protection Status/Configuration	nProtCfg, ProtStatus, nBattStatus